MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

THIRD SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG3DSEBTS200

CLINICAL CHEMISTRY AND CLINICAL MICROBIOLOGY-DSE

Time: 2 hrs Total:70 marks

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A
Fill in the blanks
Answer All Questions
Each question carries 1 mark

1. Obesity increases the risk of developing _____, diabetes, and heart disease. [K] [CO1]
2. Candidiasis is caused by----------[K][CO4][CO5]
3. AIDS is caused by the _____ virus. [K] [CO1]
4. The hormone insulin is secreted by the _____ of the pancreas. [K] [CO2][CO3]
5. Rhino viruses cause----------[K][CO4][CO5]
6. An underactive thyroid gland leads to a condition called _____.[U] [CO2][CO3]
7. ----------method is used for the estimation of serum urea[K][CO3]
8. Normal value of serum uric acidis----------[K][CO3]
9. Elevated levels of serum creatinine indicate a possible disorder in the _____.[An][CO2][CO3]
10. In chronic kidney disease, patients are advised to limit _____ intake to reduce uremic symptoms.[A][CO2][CO3]**[1X10=10 marks]**

Part B
Short questions Type Questions
Answer 10 Questions
Each question carries3 marks

(Answer any 10 questions)

11. Write any three biochemical parameters used to assess kidney function. [K][CO1] [CO2]
12. Discuss the factors that can trigger candidiasis [U][CO4]
13. How does obesity increase the risk of cardiovascular disease? [U][CO1]
14. Write down the clinical features of RSV[U][CO4][CO5]
15. A patient with elevated ALT and AST values is likely to have a disorder in which organ? Justify your answer[A][CO2][CO3]

16. What is the underlying principle of acid-fast staining? [U][CO4][CO5]
17. Suggest a basic biochemical test to evaluate kidney damage and explain its significance.[A][CO2][CO3]
18. Comment on Aspergilloma [U][CO4]
19. Mention three major symptoms of diabetes mellitus.[K][CO1]
20. Describe the selective and differential media used for isolating and identifying Salmonella, highlighting their key characteristics. [U][CO4][CO5]
21. Analyze the relationship between liver function and bilirubin levels.[An][CO2][CO3]
22. What are the recommended strategies for preventing and treating malaria, including the prophylactic measures? [U][CO4][CO5**] [3X10=30 marks]**

Part C

Short Essay Type Questions

Answer6 Questions
Each question carries 5 marks

23. Write a short note on the biochemical function of the liver.[K][CO1]
24. Explain the mode of transmission and pathogenicity of Filariasis[U][CO4][CO5]
25. A person has elevated TSH and low T3/T4 levels.suggest a diagnosis and treatment plan.[A][CO2][CO3]
26. How does Vibrio cholerae infect the host and cause cholera? [U][CO4][CO5]
27. Explain the role of the kidney in maintaining homeostasis.[K][CO1]
28. Explain the laboratory diagnosis of enteric fever [U][CO4][CO5]
29. Compare alcoholic liver disease and non-alcoholic fatty liver disease in terms of causes and biochemical features.[An][CO1]
30. Describe the mechanisms by which Corynebacterium diphtheriae causes disease, focusing on the role of diphtheria toxin[U][CO4][CO5**][5X6=30 marks]**

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